Phase2-Week5-Day-2

Class work :

HR-Resume-Implementation-Phasebyphase

create an empty folder named "HR Resume portal"
paste the "Resume_Ingestion_Architecture_Phasewise.md " file inside the empty workspace

Use this prompt as reference for implementation

Refer the architecture implementation file and do the implementation phase by phase .

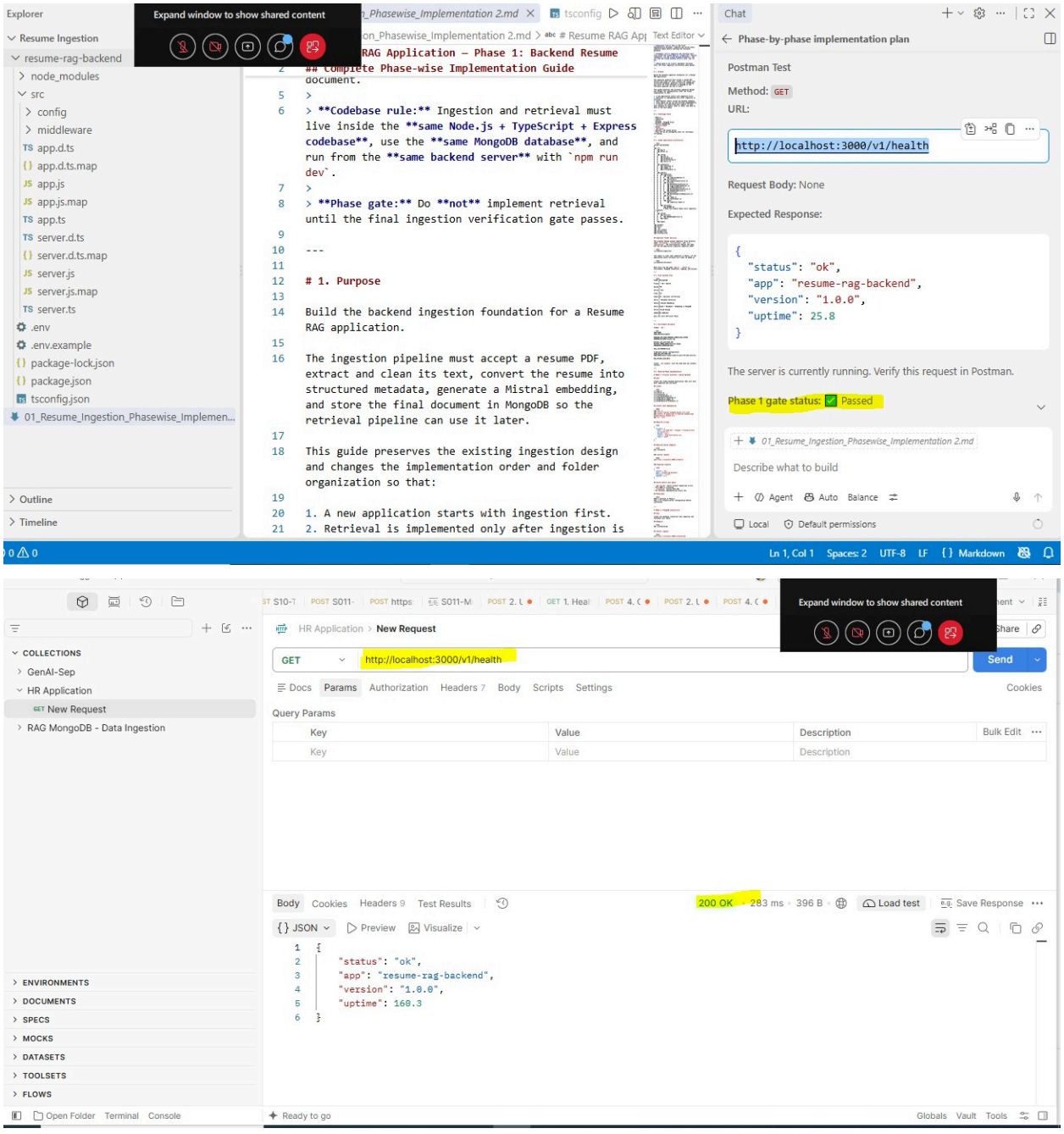
Once one phase is completed,proceed to the next phase.
[STRICT] DO NOT PROCEED TO NEXT PHASE UNTIL I APPROVE.
[STRICT]DO NOT HALLUCINATE AND DO NOT MISS ANY CONTEXT FROM THE ARCHITECTURE.GO WITH IN SCOPE OF ARCHITECTURE.
[MANDATORY]AFTER EVERY PHASE ,I AM EXPECTING REQUEST BODY AND RESPONSE BODY TO TEST IN POSTMAN

Note :Each phase should be tested in postman and once you get 200 ok status proceed to next phase

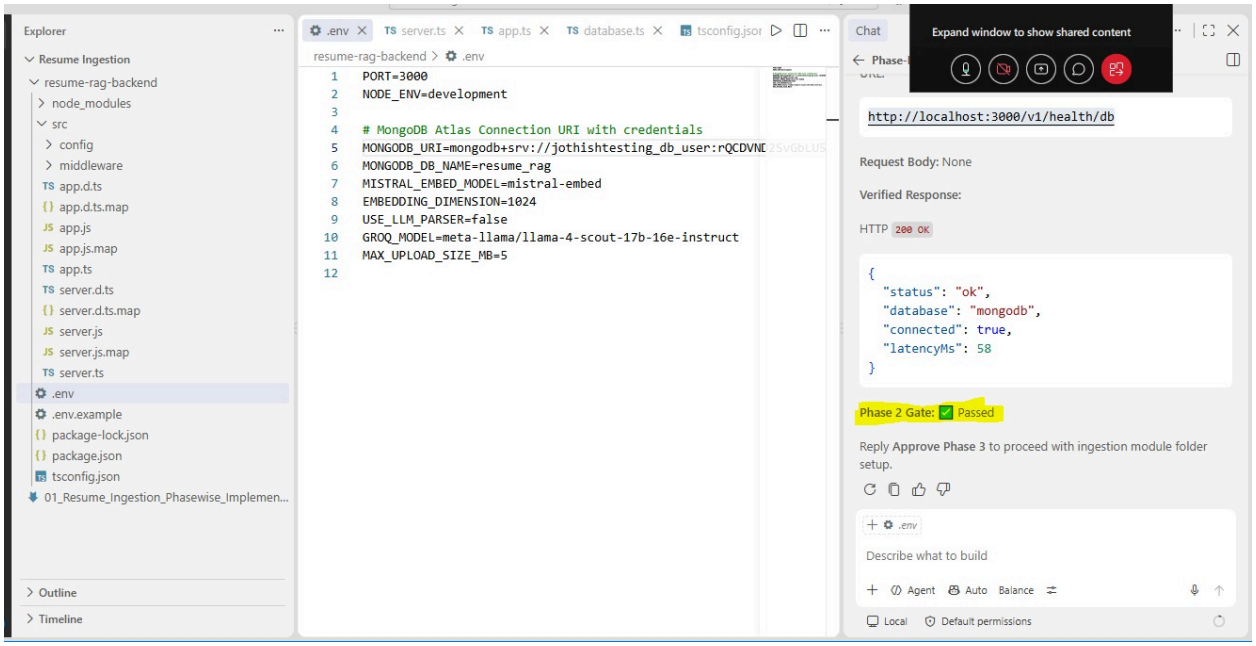
If any errors,copy the error and fix it with the help of Copilot or any IDE and please make sure your server is up and running

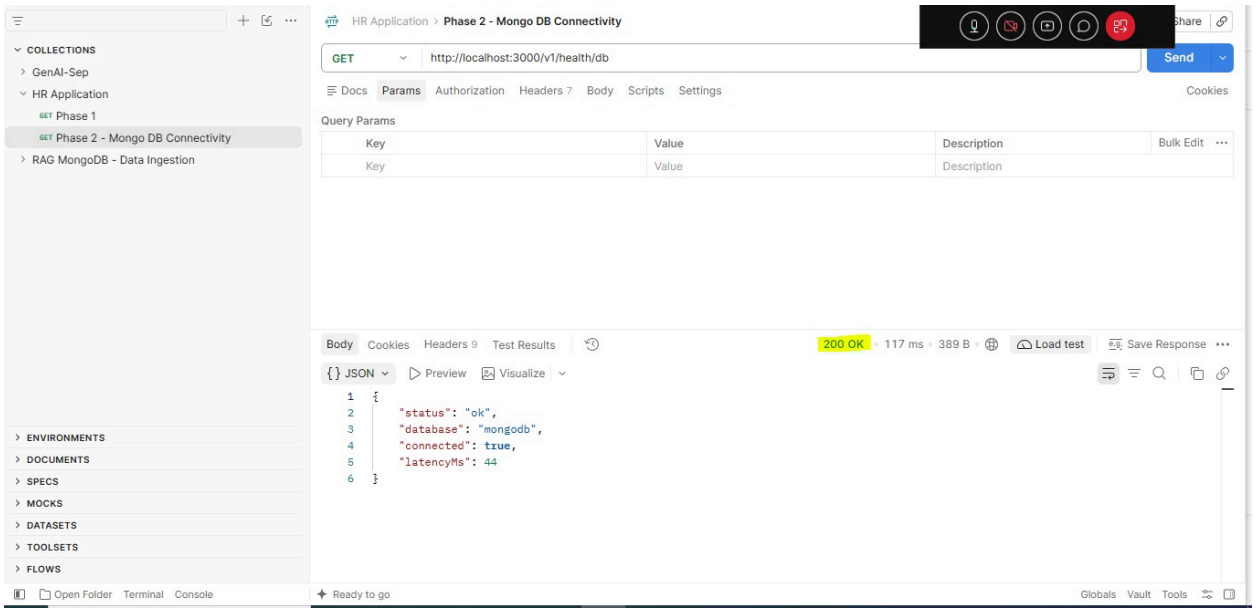
Create a separate collection for Resume storage before going into phase 11.

Phase 1:

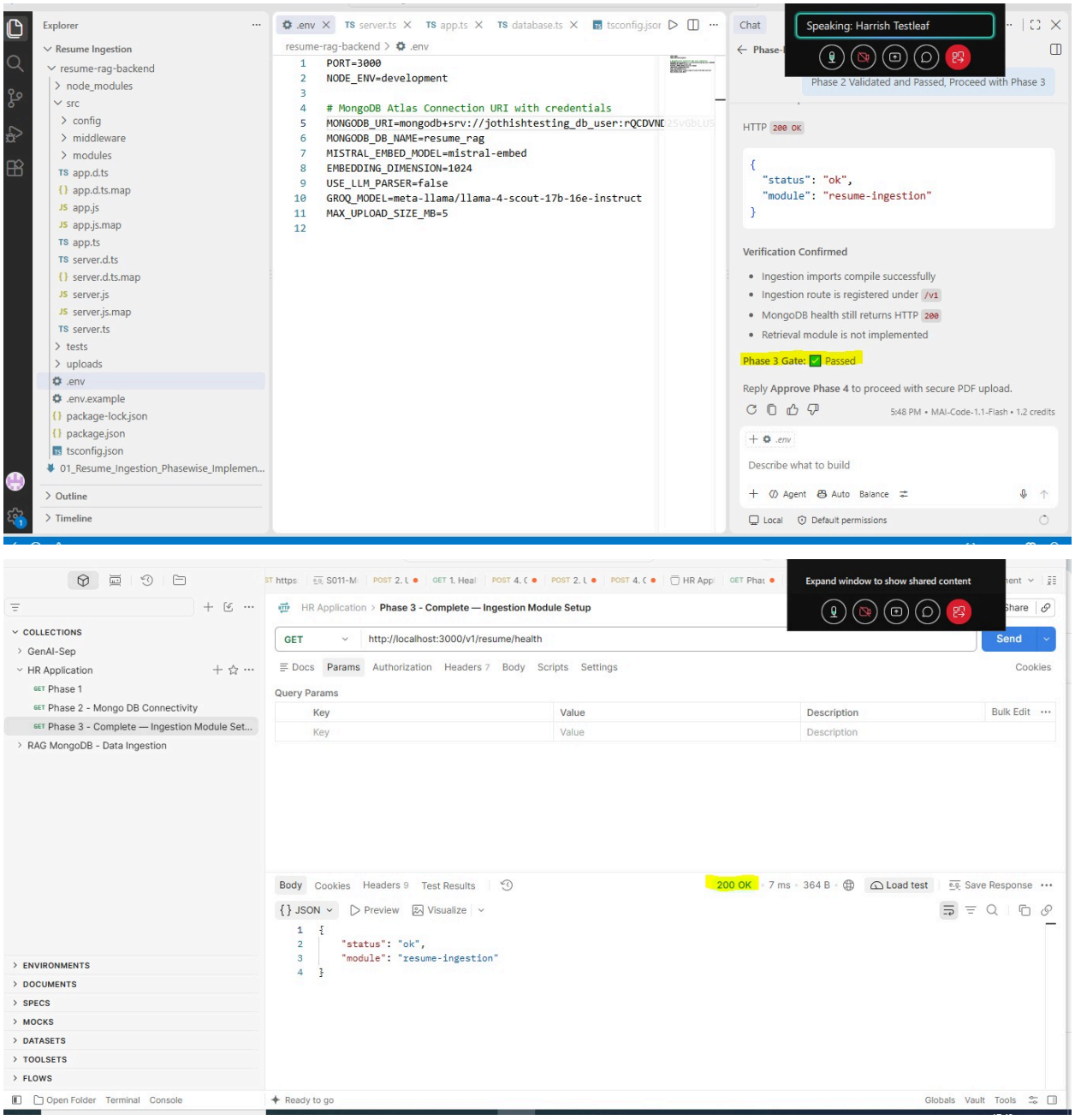


Phase 2:

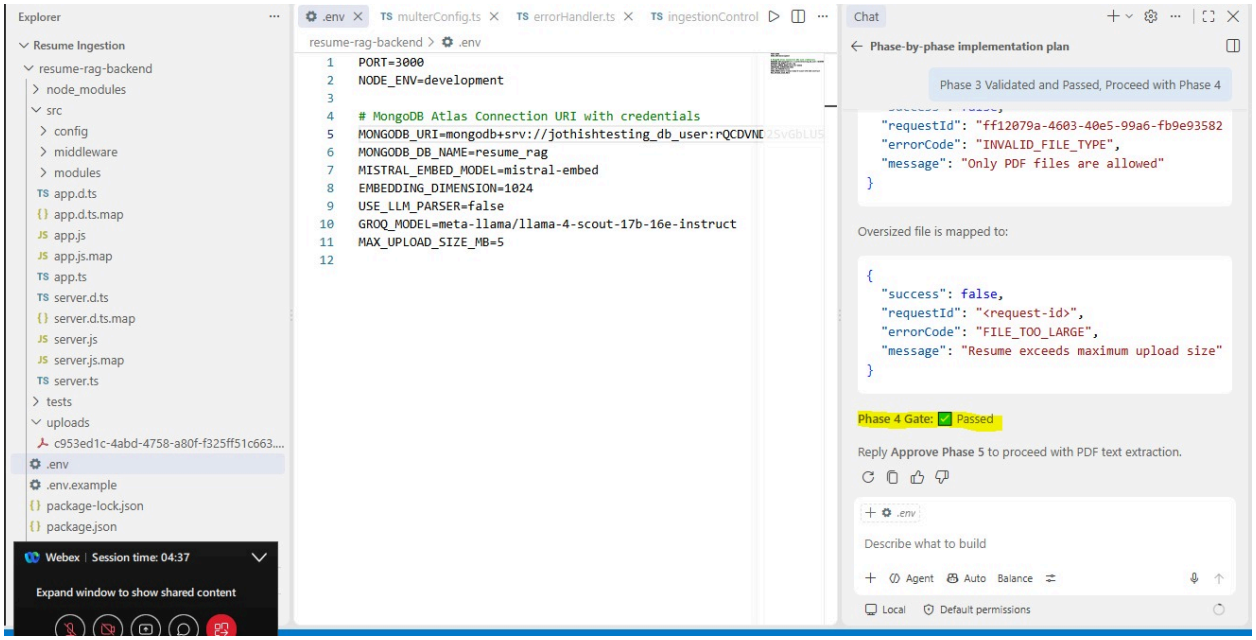

Explorer
Resume Ingestion
resume-rag-backend
node_modules
src
config
middleware
app.d.ts
app.d.ts.map
app.js
app.js.map
app.ts
server.d.ts
server.d.ts.map
server.js
server.js.map
server.ts
.env
.env.example
package-lock.json
package.json
tsconfig.json
01_Resume_Ingestion_Phasewise_Implemen...
Outline
Timeline
.env
server.ts
app.ts
database.ts
tsconfig.jsor
resume-rag-backend > .env
PORT=3000
NODE_ENV=development
MongoDB Atlas Connection URI with credentials
MONGODB_URI=mongodb+srv://jothishtesting_db_user:rQCDVNI
MONGODB_DB_NAME=resume_rag
MISTRAL_EMBED_MODEL=mistral-embed
EMBEDDING_DIMENSION=1024
USE_LLM_PARSER=false
GROQ_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
MAX_UPLOAD_SIZE_MB=5
Chat
Expand window to show shared content
http://localhost:3000/v1/health/db
Request Body: None
Verified Response:
HTTP 200 OK
{
"status": "ok",
"database": "mongodb",
"connected": true,
"latencyMs": 58
}
Phase 2 Gate: Passed
Reply Approve Phase 3 to proceed with ingestion module folder setup.
Describe what to build
Agent
Auto
Balance
Local
Default permissions

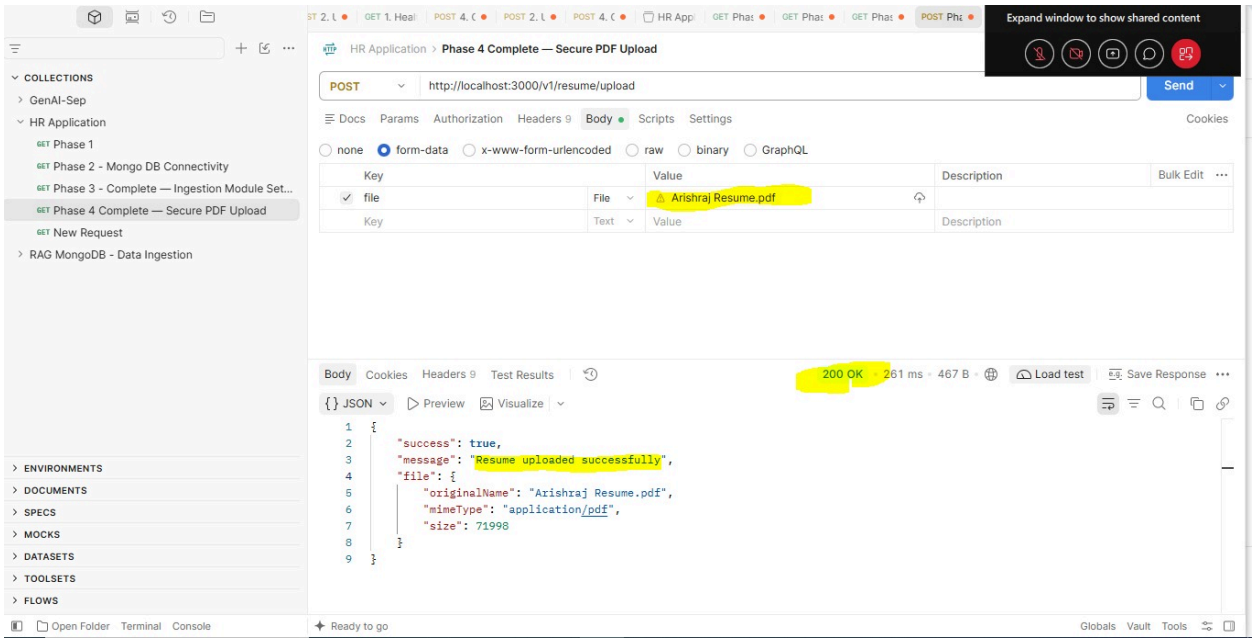

COLLECTIONS
GenAI-Sep
HR Application
Phase 1
Phase 2 - Mongo DB Connectivity
RAG MongoDB - Data Ingestion
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Open Folder
Terminal
Console
HR Application > Phase 2 - Mongo DB Connectivity
GET
http://localhost:3000/v1/health/db
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers 9
Test Results
200 OK
117 ms
389 B
Load test
Save Response
JSON
Preview
Visualize
{
"status": "ok",
"database": "mongodb",
"connected": true,
"latencyMs": 44
}
Ready to go
Globals
Vault
Tools

Phase 3:

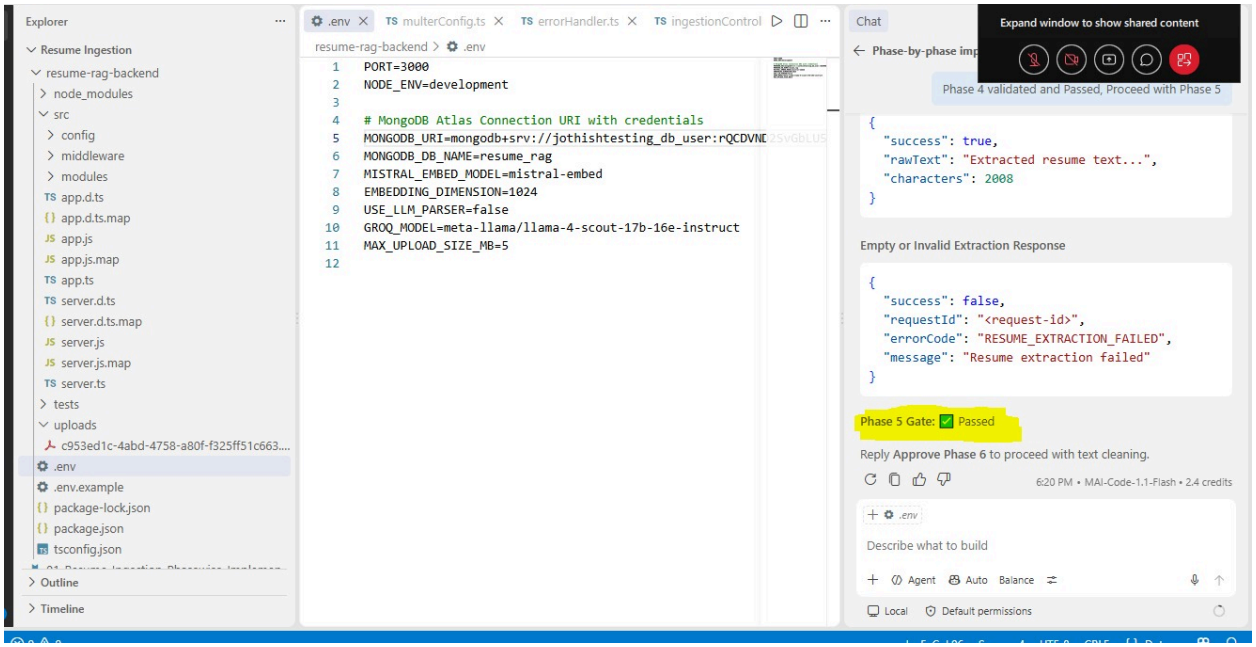

Explorer
Resume Ingestion
resume-rag-backend
node_modules
src
config
middleware
modules
app.d.ts
app.d.ts.map
app.js
app.js.map
app.ts
server.d.ts
server.d.ts.map
server.js
server.js.map
server.ts
tests
uploads
.env
.env.example
package-lock.json
package.json
tsconfig.json
01_Resume_Ingestion_Phasewise_Implemen...
Outline
Timeline
.env
server.ts
app.ts
database.ts
tsconfig.jsor
resume-rag-backend > .env
PORT=3000
NODE_ENV=development
MongoDB Atlas Connection URI with credentials
MONGODB_URI=mongodb+srv://jothishtesting_db_user:rQCDVNE
MONGODB_DB_NAME=resume_rag
MISTRAL_EMBED_MODEL=mistral-embed
EMBEDDING_DIMENSION=1024
USE_LLM_PARSER=false
GROQ_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
MAX_UPLOAD_SIZE_MB=5
Chat
Speaking: Harrish Testleaf
Phase 2 Validated and Passed, Proceed with Phase 3
HTTP 200 OK
{
"status": "ok",
"module": "resume-ingestion"
}
Verification Confirmed
Ingestion imports compile successfully
Ingestion route is registered under /v1
MongoDB health still returns HTTP 200
Retrieval module is not implemented
Phase 3 Gate: Passed
Reply Approve Phase 4 to proceed with secure PDF upload.
5:48 PM • MAI-Code-1.1-Flash • 1.2 credits
.env
Describe what to build
Agent
Auto
Balance
Local
Default permissions
Expand window to show shared content
COLLECTIONS
GenAI-Sep
HR Application
Phase 1
Phase 2 - Mongo DB Connectivity
Phase 3 - Complete — Ingestion Module Set...
RAG MongoDB - Data Ingestion
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
HR Application > Phase 3 - Complete — Ingestion Module Setup
GET
http://localhost:3000/v1/resume/health
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers 9
Test Results
200 OK
7 ms
364 B
Load test
Save Response
{ } JSON
Preview
Visualize
{
"status": "ok",
"module": "resume-ingestion"
}
Open Folder
Terminal
Console
Ready to go
Globals
Vault
Tools

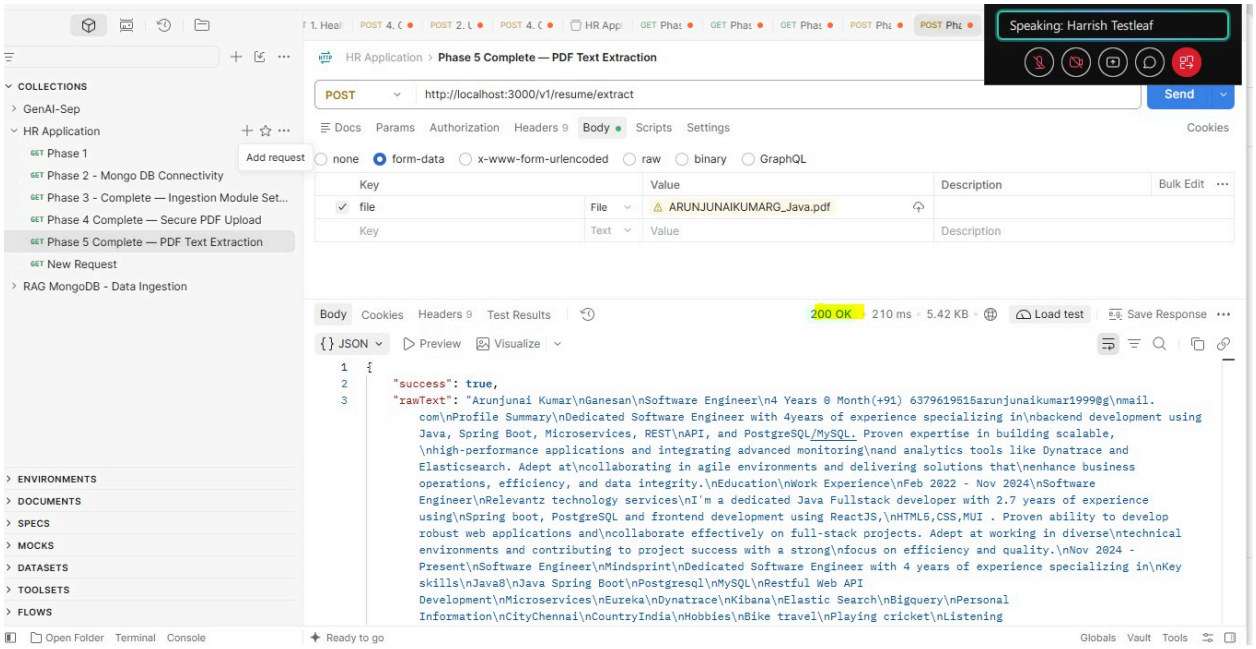
Phase 4:


Explorer
Resume Ingestion
resume-rag-backend
node_modules
src
config
middleware
modules
app.d.ts
app.d.ts.map
app.js
app.js.map
app.ts
server.d.ts
server.d.ts.map
server.js
server.js.map
server.ts
tests
uploads
c953ed1c-4abd-4758-a80f-f325ff51c663....
.env
.env.example
package-lock.json
package.json
Webex | Session time: 04:37
Expand window to show shared content
.env
multerConfig.ts
errorHandler.ts
ingestionControl
resume-rag-backend > .env
PORT=3000
NODE_ENV=development
MongoDB Atlas Connection URI with credentials
MONGODB_URI=mongodb+srv://jothishtesting_db_user:rQCDVNE
MONGODB_DB_NAME=resume_rag
MISTRAL_EMBED_MODEL=mistral-embed
EMBEDDING_DIMENSION=1024
USE_LLM_PARSER=false
GROQ_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
MAX_UPLOAD_SIZE_MB=5
Chat
Phase-by-phase implementation plan
Phase 3 Validated and Passed, Proceed with Phase 4
"requestId": "ff12079a-4603-40e5-99a6-fb9e93582
"errorCode": "INVALID_FILE_TYPE",
"message": "Only PDF files are allowed"
Oversized file is mapped to:
"success": false,
"requestId": "<request-id>",
"errorCode": "FILE_TOO_LARGE",
"message": "Resume exceeds maximum upload size"
Phase 4 Gate: ✅ Passed
Reply Approve Phase 5 to proceed with PDF text extraction.
Describe what to build
Agent
Auto
Balance
Local
Default permissions

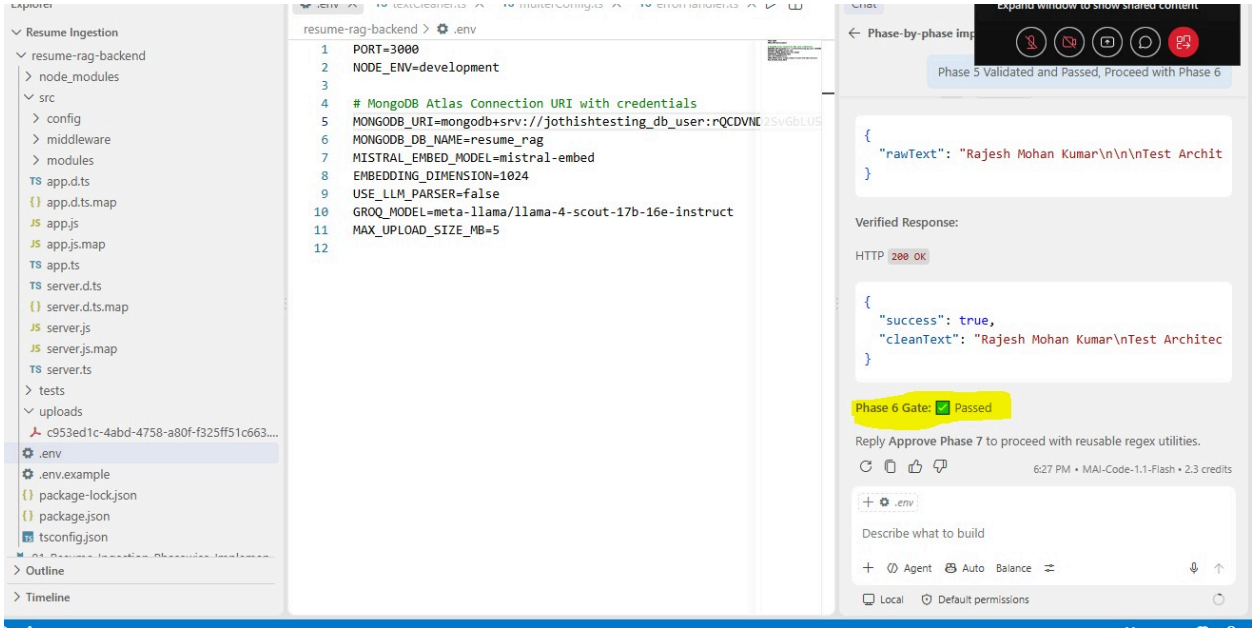

Expand window to show shared content
COLLECTIONS
GenAI-Sep
HR Application
Phase 1
Phase 2 - Mongo DB Connectivity
Phase 3 - Complete — Ingestion Module Set...
Phase 4 Complete — Secure PDF Upload
New Request
RAG MongoDB - Data Ingestion
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
HR Application > Phase 4 Complete — Secure PDF Upload
POST
http://localhost:3000/v1/resume/upload
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Key
Value
Description
Bulk Edit
file
File
Arishraj Resume.pdf
Text
Body
Cookies
Headers 9
Test Results
200 OK
261 ms
467 B
Load test
Save Response
JSON
Preview
Visualize
{
"success": true,
"message": "Resume uploaded successfully",
"file": {
"originalName": "Arishraj Resume.pdf",
"mimeType": "application/pdf",
"size": 71998
}
}
Open Folder
Terminal
Console
Ready to go
Globals
Vault
Tools

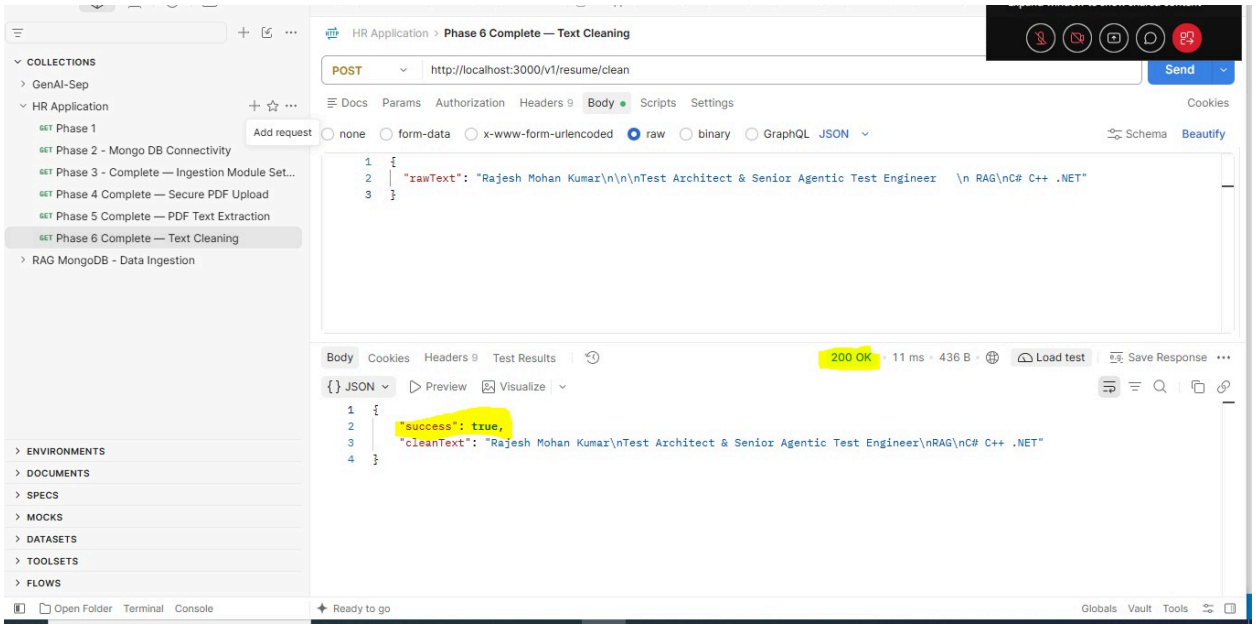
Phase 5:



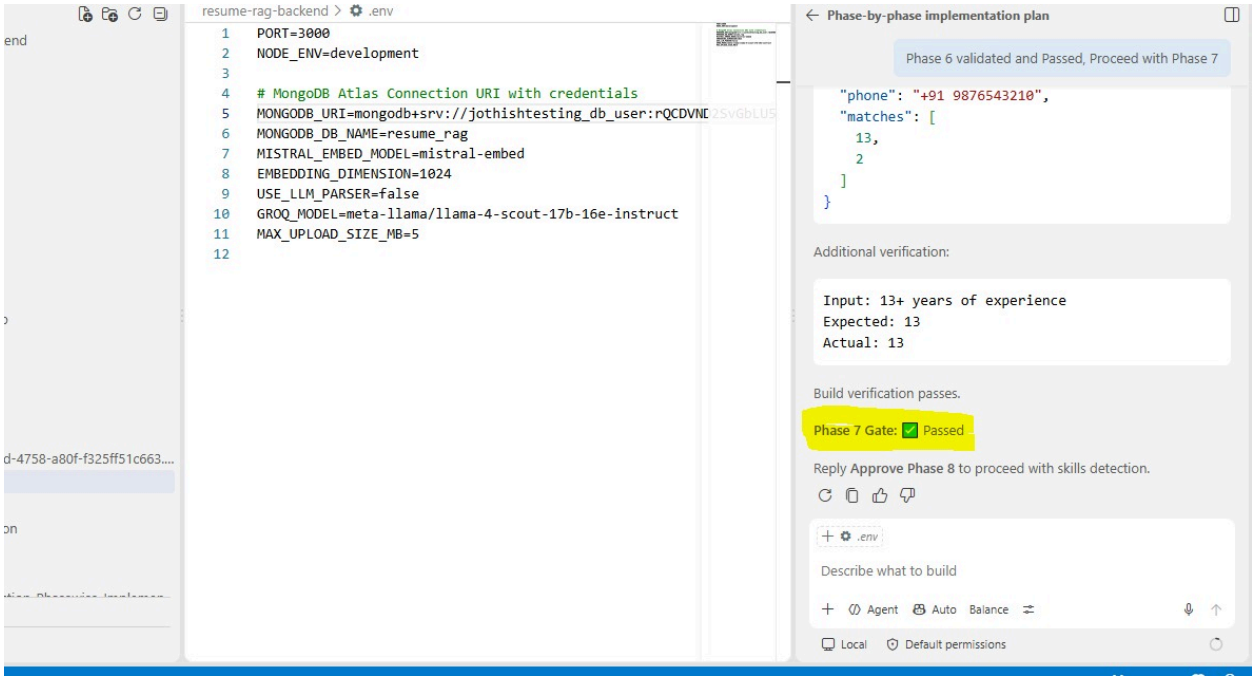


Phase 6:

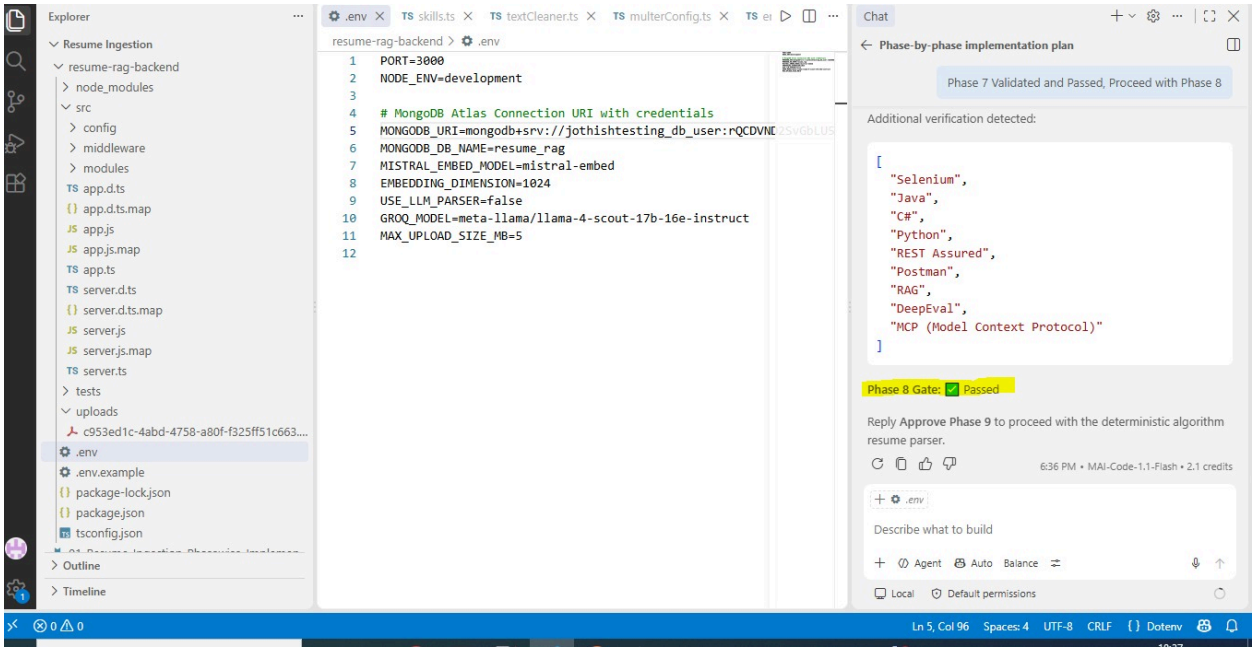

Resume Ingestion
resume-rag-backend
node_modules
src
config
middleware
modules
app.d.ts
app.d.ts.map
app.js
app.js.map
app.ts
server.d.ts
server.d.ts.map
server.js
server.js.map
server.ts
tests
uploads
.env
.env.example
package-lock.json
package.json
tsconfig.json
Outline
Timeline
resume-rag-backend > .env
PORT=3000
NODE_ENV=development
MongoDB Atlas Connection URI with credentials
MONGODB_URI=mongodb+srv://jothishtesting_db_user:rQCDVNE
MONGODB_DB_NAME=resume_rag
MISTRAL_EMBED_MODEL=mistral-embed
EMBEDDING_DIMENSION=1024
USE_LLM_PARSER=false
GROQ_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
MAX_UPLOAD_SIZE_MB=5
Phase 5 Validated and Passed, Proceed with Phase 6
"rawText": "Rajesh Mohan Kumar\n\n\nTest Archit
Verified Response:
HTTP 200 OK
"success": true,
"cleanText": "Rajesh Mohan Kumar\nTest Architec
Phase 6 Gate: Passed
Reply Approve Phase 7 to proceed with reusable regex utilities.
6:27 PM • MAI-Code-1.1-Flash • 2.3 credits
Describe what to build
Agent
Auto
Balance
Local
Default permissions

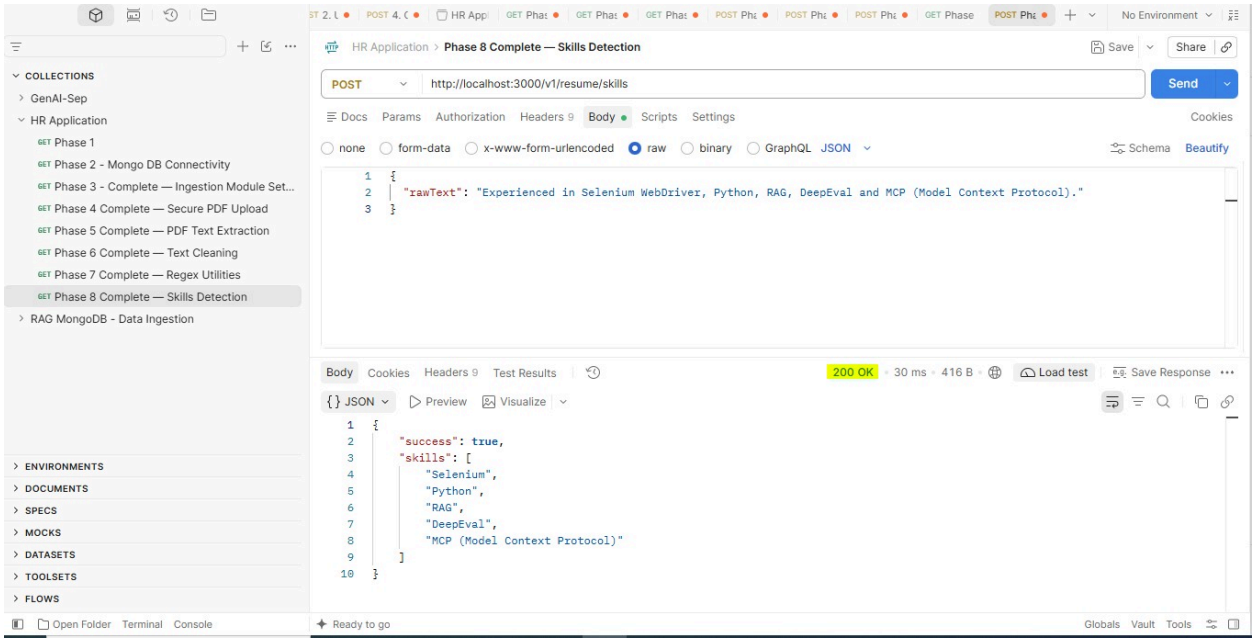

COLLECTIONS
GenAI-Sep
HR Application
Phase 1
Phase 2 - Mongo DB Connectivity
Phase 3 - Complete — Ingestion Module Set...
Phase 4 Complete — Secure PDF Upload
Phase 5 Complete — PDF Text Extraction
Phase 6 Complete — Text Cleaning
RAG MongoDB - Data Ingestion
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
HR Application > Phase 6 Complete — Text Cleaning
POST
http://localhost:3000/v1/resume/clean
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"rawText": "Rajesh Mohan Kumar\n\n\nTest Architect & Senior Agentic Test Engineer \n RAG\nC# C++ .NET"
}
Body
Cookies
Headers 9
Test Results
200 OK
11 ms
436 B
Load test
Save Response
{} JSON
Preview
Visualize
{
"success": true,
"cleanText": "Rajesh Mohan Kumar\nTest Architect & Senior Agentic Test Engineer\nRAG\nC# C++ .NET"
}
Open Folder
Terminal
Console
Ready to go
Globals
Vault
Tools

Phase 7:

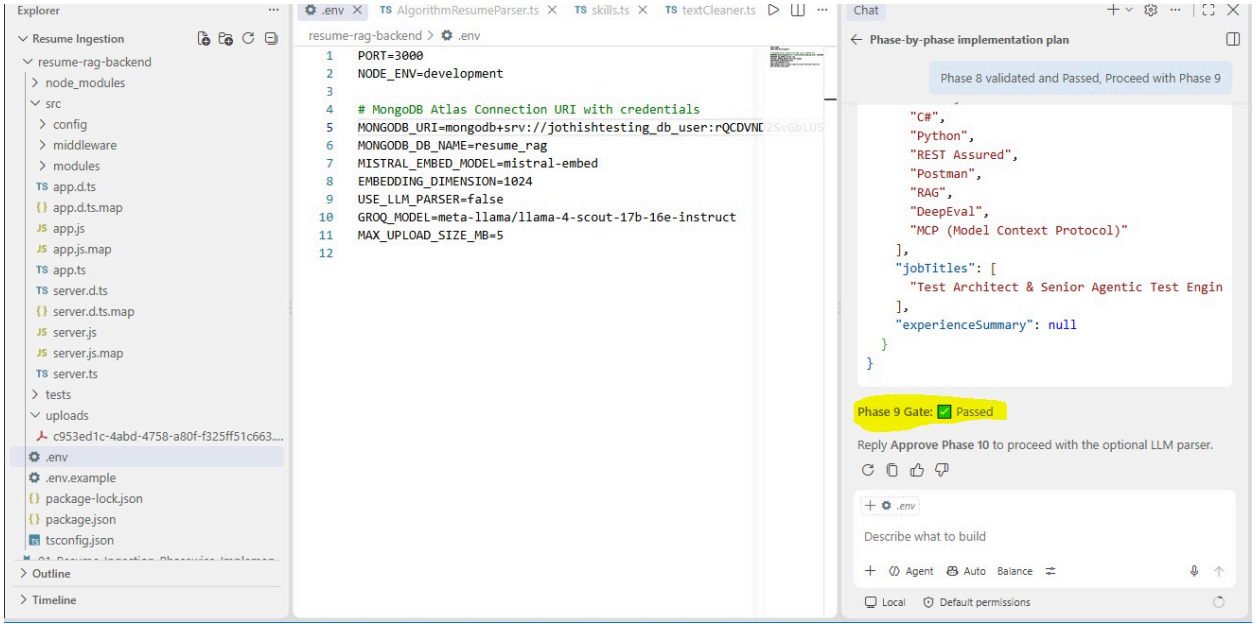

resume-rag-backend > .env
PORT=3000
NODE_ENV=development
MongoDB Atlas Connection URI with credentials
MONGODB_URI=mongodb+srv://jothishtesting_db_user:rQCDVNE
MONGODB_DB_NAME=resume_rag
MISTRAL_EMBED_MODEL=mistral-embed
EMBEDDING_DIMENSION=1024
USE_LLM_PARSER=false
GROQ_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
MAX_UPLOAD_SIZE_MB=5
Phase-by-phase implementation plan
Phase 6 validated and Passed, Proceed with Phase 7
"phone": "+91 9876543210",
"matches": [
13,
2
]
}
Additional verification:
Input: 13+ years of experience
Expected: 13
Actual: 13
Build verification passes.
Phase 7 Gate: Passed
Reply Approve Phase 8 to proceed with skills detection.
Describe what to build
Agent
Auto
Balance
Local
Default permissions

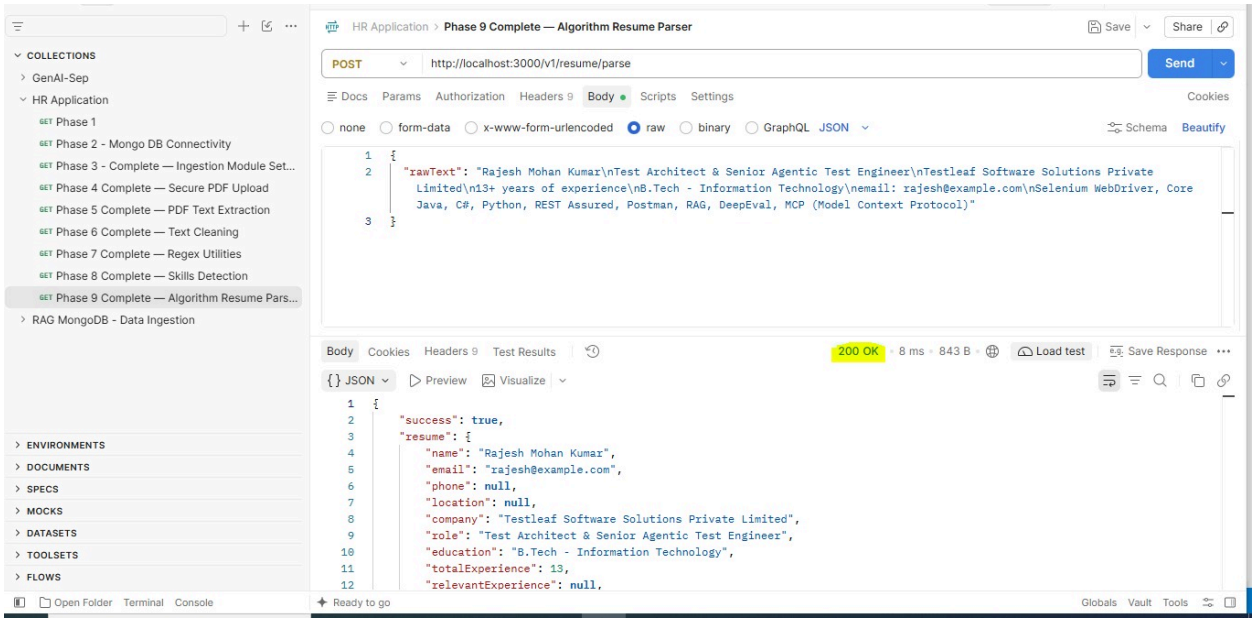
Phase 8:


Explorer
Resume Ingestion
resume-rag-backend
node_modules
src
config
middleware
modules
app.d.ts
app.d.ts.map
app.js
app.js.map
app.ts
server.d.ts
server.d.ts.map
server.js
server.js.map
server.ts
tests
uploads
.env
.env.example
package-lock.json
package.json
tsconfig.json
Outline
Timeline
resume-rag-backend > .env
PORT=3000
NODE_ENV=development
MongoDB Atlas Connection URI with credentials
MONGODB_URI=mongodb+srv://jothishtesting_db_user:rQCDVNE
MONGODB_DB_NAME=resume_rag
MISTRAL_EMBED_MODEL=mistral-embed
EMBEDDING_DIMENSION=1024
USE_LLM_PARSER=false
GROQ_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
MAX_UPLOAD_SIZE_MB=5
Chat
Phase-by-phase implementation plan
Phase 7 Validated and Passed, Proceed with Phase 8
Additional verification detected:
[
"Selenium",
"Java",
"C#",
"Python",
"REST Assured",
"Postman",
"RAG",
"DeepEval",
"MCP (Model Context Protocol)"
]
Phase 8 Gate: Passed
Reply Approve Phase 9 to proceed with the deterministic algorithm resume parser.
6:36 PM • MAI-Code-1.1-Flash • 2.1 credits
Describe what to build
Agent
Auto
Balance
Local
Default permissions
Ln 5, Col 96
Spaces: 4
UTF-8
CRLF
Dotenv

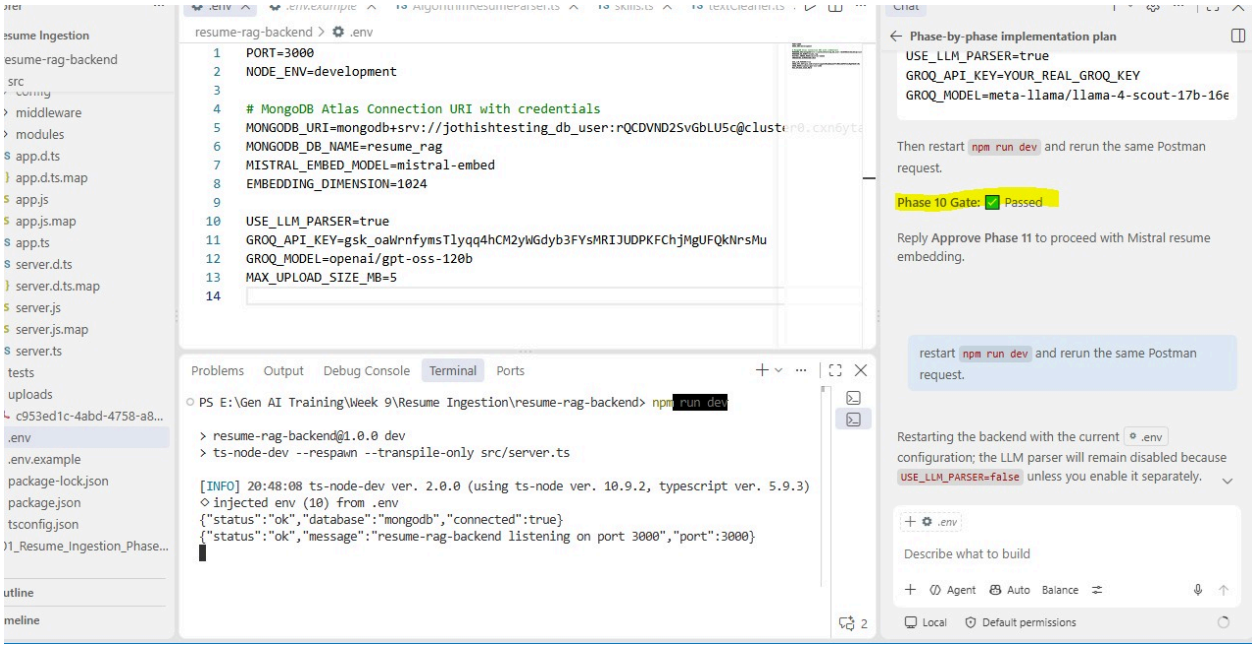

HR Application > Phase 8 Complete — Skills Detection
POST
http://localhost:3000/v1/resume/skills
Send
{
"rawText": "Experienced in Selenium WebDriver, Python, RAG, DeepEval and MCP (Model Context Protocol)."
}
200 OK
{
"success": true,
"skills": [
"Selenium",
"Python",
"RAG",
"DeepEval",
"MCP (Model Context Protocol)"
]
}

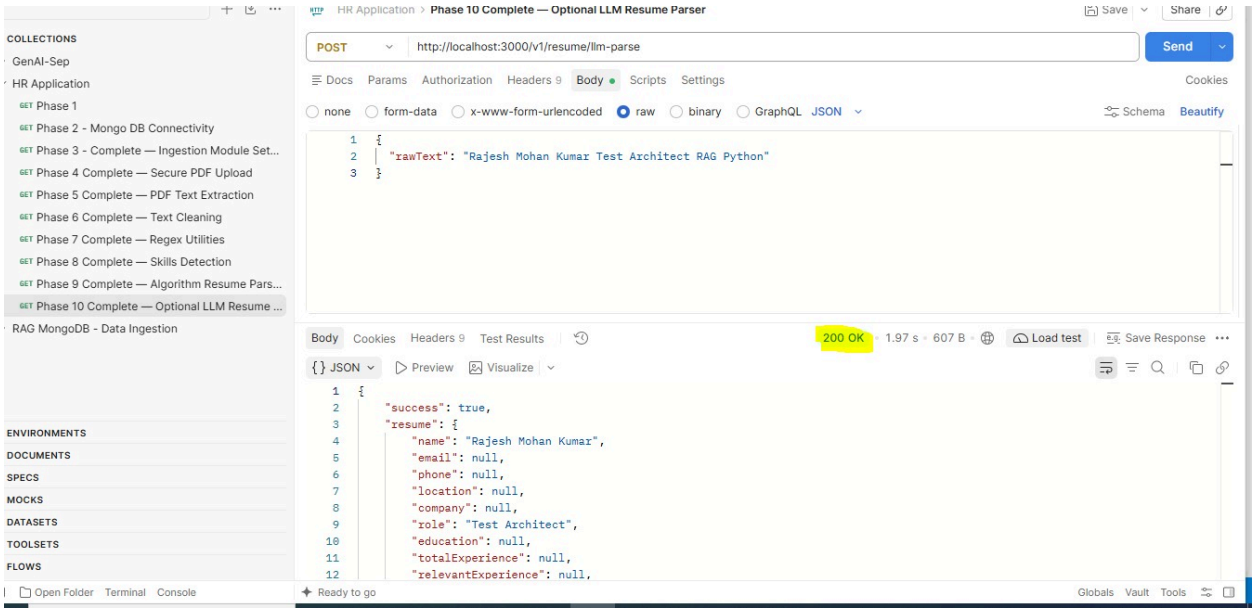
Phase 9:


PORT=3000
NODE_ENV=development
MongoDB Atlas Connection URI with credentials
MONGODB_DB_NAME=resume_rag
MISTRAL_EMBED_MODEL=mistral-embed
EMBEDDING_DIMENSION=1024
USE_LLM_PARSER=false
GROQ_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
MAX_UPLOAD_SIZE_MB=5
Phase 8 validated and Passed, Proceed with Phase 9
"C#",
"Python",
"REST Assured",
"Postman",
"RAG",
"DeepEval",
"MCP (Model Context Protocol)"
"jobTitles": [
"Test Architect & Senior Agentic Test Engin
"experienceSummary": null
Phase 9 Gate: Passed
Reply Approve Phase 10 to proceed with the optional LLM parser.
Describe what to build

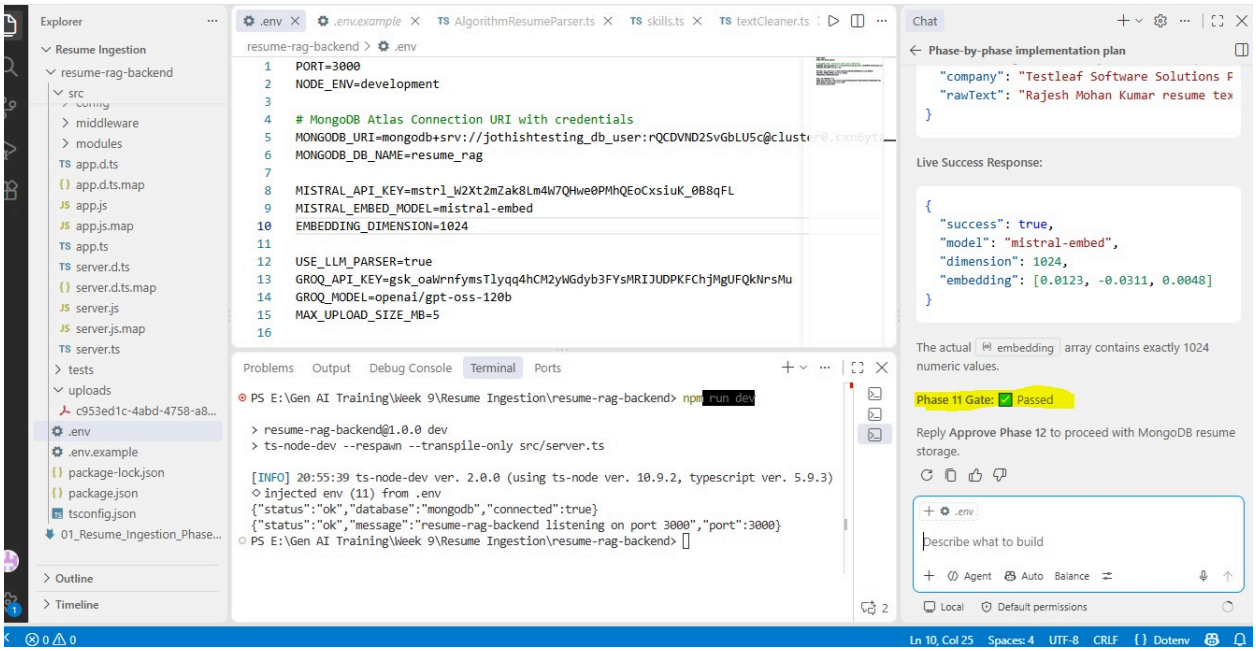

HR Application > Phase 9 Complete — Algorithm Resume Parser
Save
Share
POST
http://localhost:3000/v1/resume/parse
Send
Docs Params Authorization Headers 9 Body Scripts Settings Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
{
"rawText": "Rajesh Mohan Kumar\nTest Architect & Senior Agentic Test Engineer\nTestleaf Software Solutions Private Limited\n13+ years of experience\nB.Tech - Information Technology\nemail: rajesh@example.com\nSelenium WebDriver, Core Java, C#, Python, REST Assured, Postman, RAG, DeepEval, MCP (Model Context Protocol)"
}
Body Cookies Headers 9 Test Results
200 OK
8 ms
843 B
Load test
Save Response
{ } JSON
Preview
Visualize
{
"success": true,
"resume": {
"name": "Rajesh Mohan Kumar",
"email": "rajesh@example.com",
"phone": null,
"location": null,
"company": "Testleaf Software Solutions Private Limited",
"role": "Test Architect & Senior Agentic Test Engineer",
"education": "B.Tech - Information Technology",
"totalExperience": 13,
"relevantExperience": null,
COLLECTIONS
GenAI-Sep
HR Application
Phase 1
Phase 2 - Mongo DB Connectivity
Phase 3 - Complete — Ingestion Module Set...
Phase 4 Complete — Secure PDF Upload
Phase 5 Complete — PDF Text Extraction
Phase 6 Complete — Text Cleaning
Phase 7 Complete — Regex Utilities
Phase 8 Complete — Skills Detection
Phase 9 Complete — Algorithm Resume Pars...
RAG MongoDB - Data Ingestion
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Open Folder Terminal Console
Ready to go
Globals Vault Tools

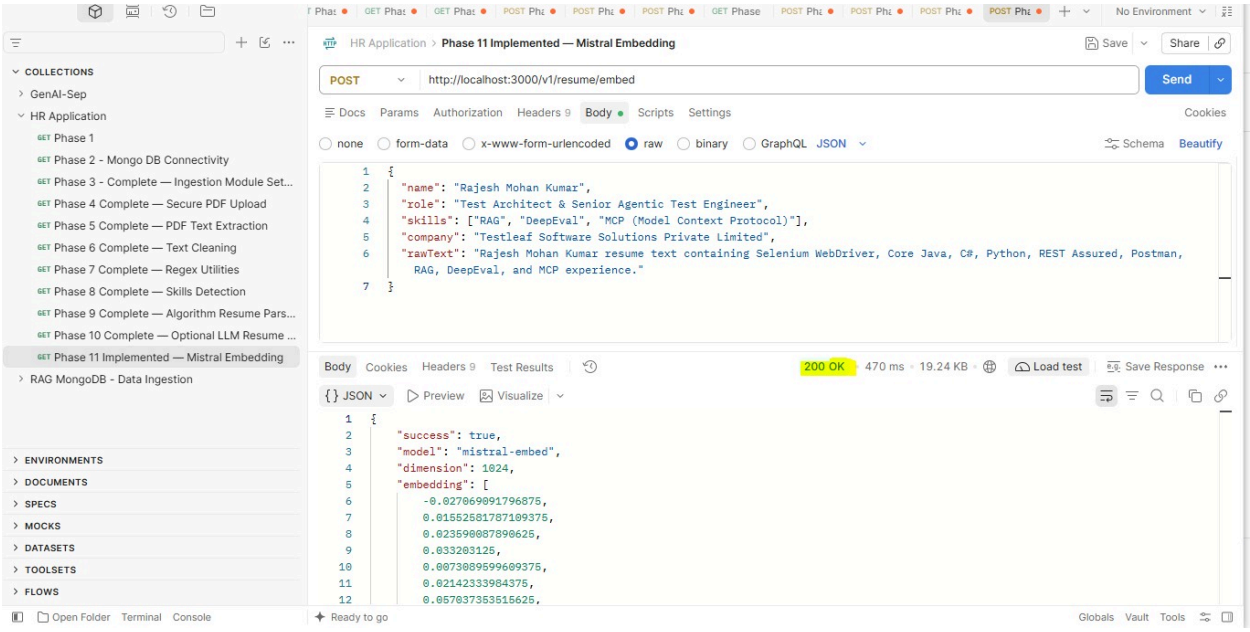
Phase 10:


resume-rag-backend > .env
PORT=3000
NODE_ENV=development
MongoDB Atlas Connection URI with credentials
MONGODB_URI=mongodb+srv://jothishtesting_db_user:rQCDVND2SvGbLU5c@cluste
MONGODB_DB_NAME=resume_rag
MISTRAL_EMBED_MODEL=mistral-embed
EMBEDDING_DIMENSION=1024
USE_LLM_PARSER=true
GROQ_API_KEY=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
GROQ_MODEL=openai/gpt-oss-120b
MAX_UPLOAD_SIZE_MB=5
Problems Output Debug Console Terminal Ports
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend> npm run dev
> resume-rag-backend@1.0.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts
[INFO] 20:48:08 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typescript ver. 5.9.3)
◇ injected env (10) from .env
{"status":"ok","database":"mongodb","connected":true}
{"status":"ok","message":"resume-rag-backend listening on port 3000","port":3000}
Phase-by-phase implementation plan
USE_LLM_PARSER=true
GROQ_API_KEY=YOUR_REAL_GROQ_KEY
GROQ_MODEL=meta-llama/llama-4-scout-17b-16e
Then restart npm run dev and rerun the same Postman request.
Phase 10 Gate: ✅ Passed
Reply Approve Phase 11 to proceed with Mistral resume embedding.
restart npm run dev and rerun the same Postman request.
Restarting the backend with the current .env configuration; the LLM parser will remain disabled because USE_LLM_PARSER=false unless you enable it separately.
Describe what to build
Agent Auto Balance
Local Default permissions
esume Ingestion
esume-rag-backend
src
middleware
modules
app.d.ts
app.d.ts.map
app.js
app.js.map
app.ts
server.d.ts
server.d.ts.map
server.js
server.js.map
server.ts
tests
uploads
.env
.env.example
package-lock.json
package.json
tsconfig.json
utline
meline

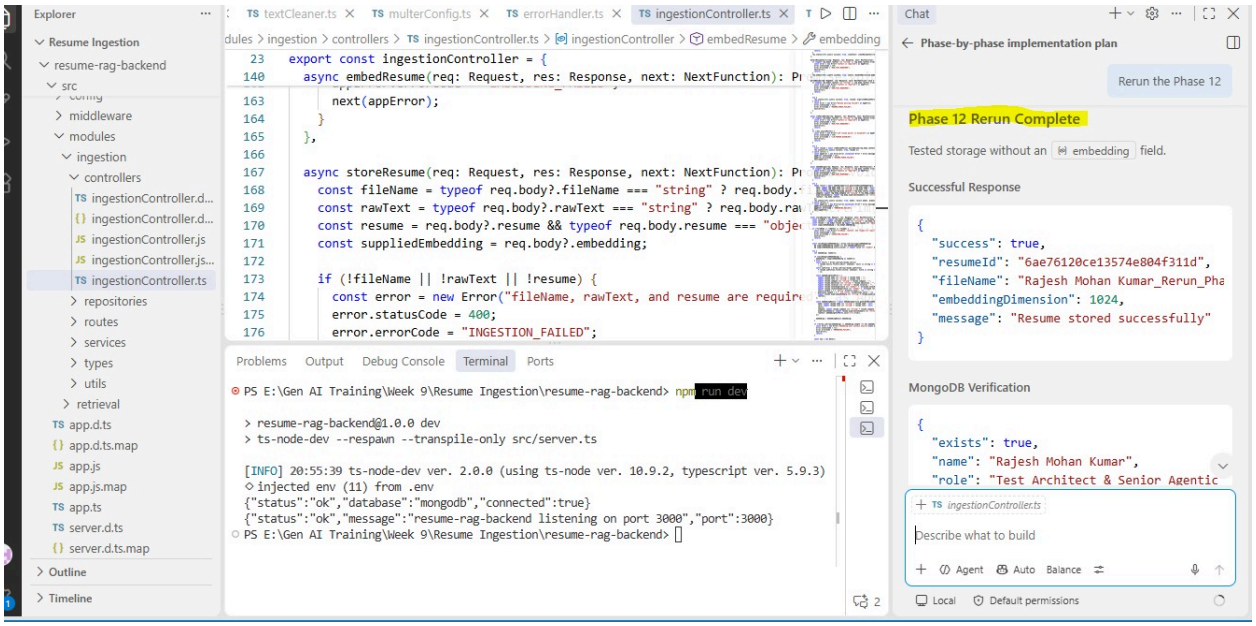

HR Application > Phase 10 Complete — Optional LLM Resume Parser
POST
http://localhost:3000/v1/resume/llm-parse
Send
{
"rawText": "Rajesh Mohan Kumar Test Architect RAG Python"
}
200 OK
{
"success": true,
"resume": {
"name": "Rajesh Mohan Kumar",
"email": null,
"phone": null,
"location": null,
"company": null,
"role": "Test Architect",
"education": null,
"totalExperience": null,
"relevantExperience": null,

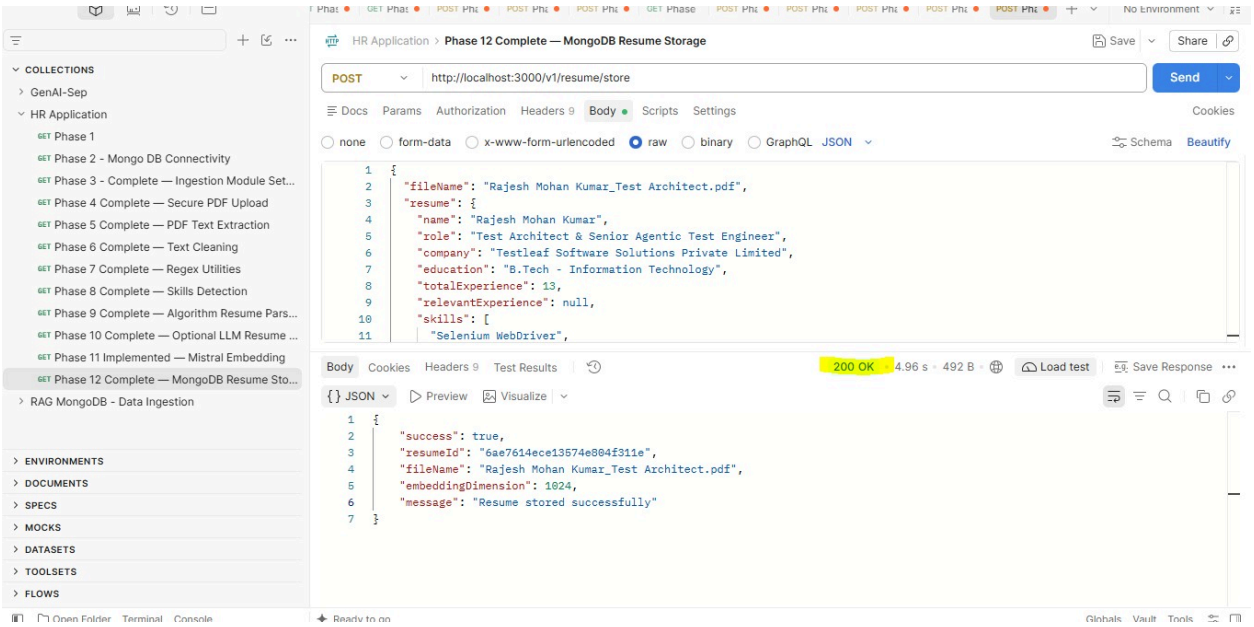
Phase 11:


PORT=3000
NODE_ENV=development
MongoDB Atlas Connection URI with credentials
MONGODB_DB_NAME=resume_rag
MISTRAL_API_KEY=mstrl_W2Xt2mZak8Lm4W7QHwe0PMhQEoCxsiuK_0B8qFL
MISTRAL_EMBED_MODEL=mistral-embed
EMBEDDING_DIMENSION=1024
USE_LLM_PARSER=true
GROQ_API_KEY=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
GROQ_MODEL=openai/gpt-oss-120b
MAX_UPLOAD_SIZE_MB=5
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend> npm run dev
> resume-rag-backend@1.0.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts
[INFO] 20:55:39 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typescript ver. 5.9.3)
injected env (11) from .env
{"status":"ok","database":"mongodb","connected":true}
{"status":"ok","message":"resume-rag-backend listening on port 3000","port":3000}
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend>
Live Success Response:
{
"success": true,
"model": "mistral-embed",
"dimension": 1024,
"embedding": [0.0123, -0.0311, 0.0048]
}
The actual embedding array contains exactly 1024 numeric values.
Phase 11 Gate: Passed
Reply Approve Phase 12 to proceed with MongoDB resume storage.

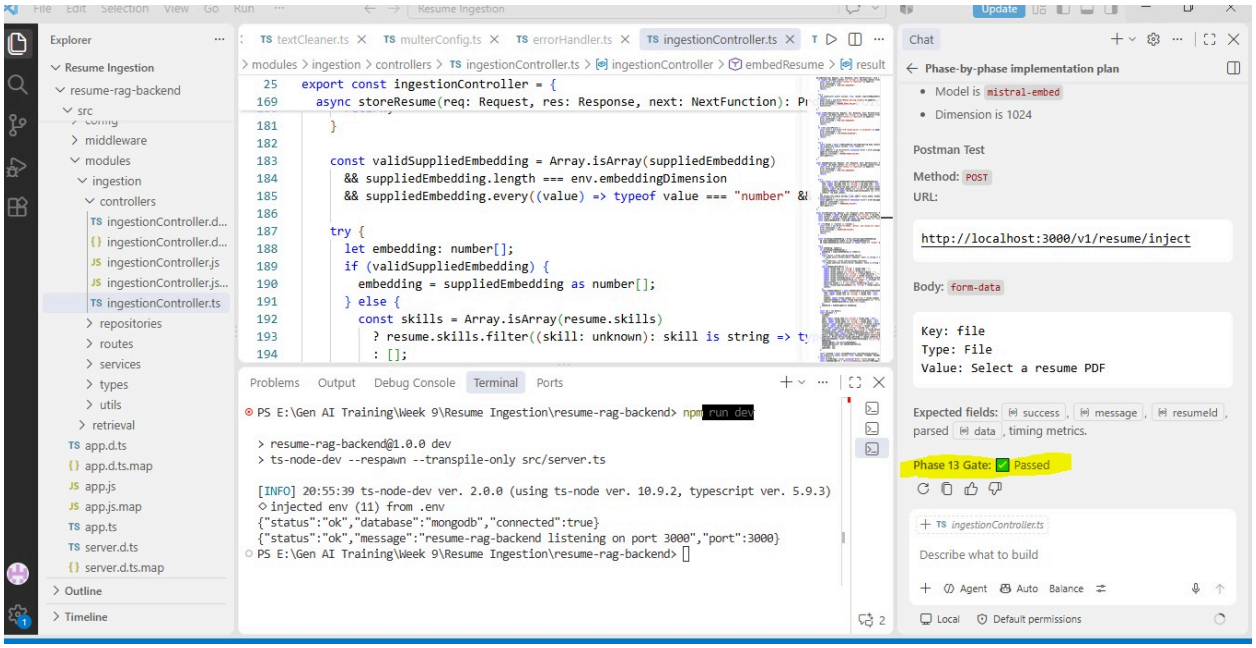

HR Application > Phase 11 Implemented — Mistral Embedding
POST
http://localhost:3000/v1/resume/embed
Send
{
"name": "Rajesh Mohan Kumar",
"role": "Test Architect & Senior Agentic Test Engineer",
"skills": ["RAG", "DeepEval", "MCP (Model Context Protocol)"],
"company": "Testleaf Software Solutions Private Limited",
"rawText": "Rajesh Mohan Kumar resume text containing Selenium WebDriver, Core Java, C#, Python, REST Assured, Postman, RAG, DeepEval, and MCP experience."
}
200 OK
470 ms
19.24 KB
{
"success": true,
"model": "mistral-embed",
"dimension": 1024,
"embedding": [
-0.027069091796875,
0.01552581787109375,
0.023590087890625,
0.033203125,
0.0073089599609375,
0.02142333984375,
0.057037353515625,

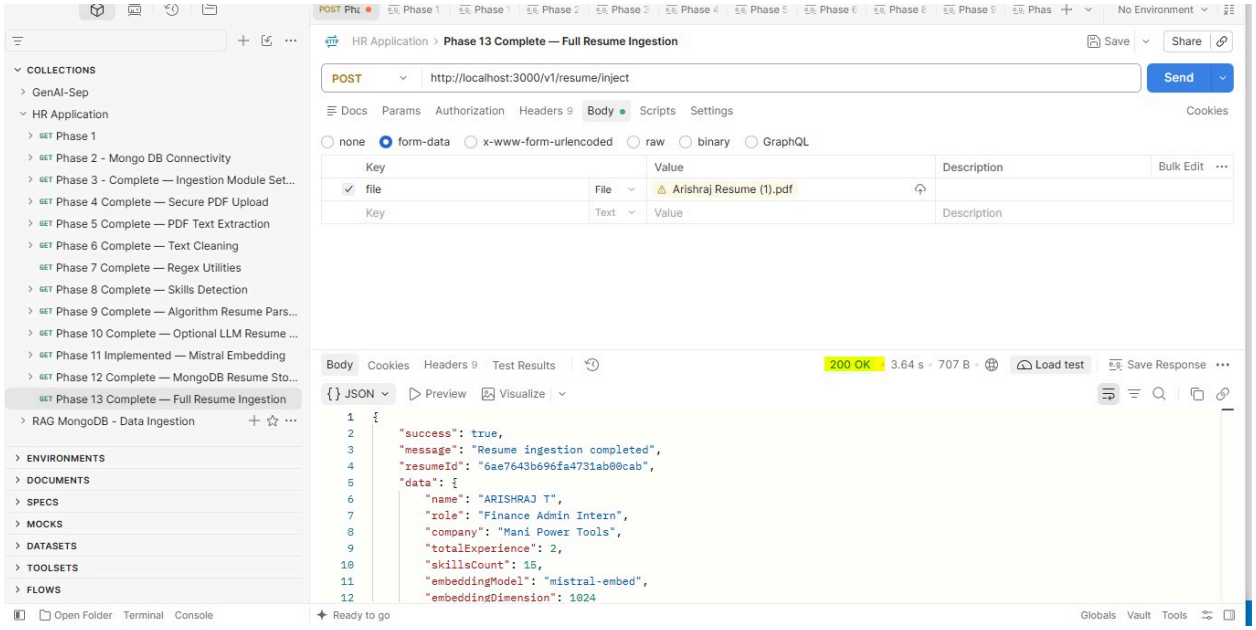
Phase 12:


export const ingestionController = {
async embedResume(req: Request, res: Response, next: NextFunction): Pr
next(appError);
},
async storeResume(req: Request, res: Response, next: NextFunction): Pr
const fileName = typeof req.body?.fileName === "string" ? req.body.
const rawText = typeof req.body?.rawText === "string" ? req.body.ra
const resume = req.body?.resume && typeof req.body.resume === "obje
const suppliedEmbedding = req.body?.embedding;
if (!fileName || !rawText || !resume) {
const error = new Error("fileName, rawText, and resume are requir
error.statusCode = 400;
error.errorCode = "INGESTION_FAILED";
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend> npm run dev
> resume-rag-backend@1.0.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts
[INFO] 20:55:39 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typescript ver. 5.9.3)
injected env (11) from .env
{"status":"ok","database":"mongodb","connected":true}
{"status":"ok","message":"resume-rag-backend listening on port 3000","port":3000}
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend>
Phase-by-phase implementation plan
Rerun the Phase 12
Phase 12 Rerun Complete
Tested storage without an embedding field.
Successful Response
{
"success": true,
"resumeId": "6ae76120ce13574e804f311d",
"fileName": "Rajesh Mohan Kumar_Rerun_Pha
"embeddingDimension": 1024,
"message": "Resume stored successfully"
}
MongoDB Verification
{
"exists": true,
"name": "Rajesh Mohan Kumar",
"role": "Test Architect & Senior Agentic
Describe what to build

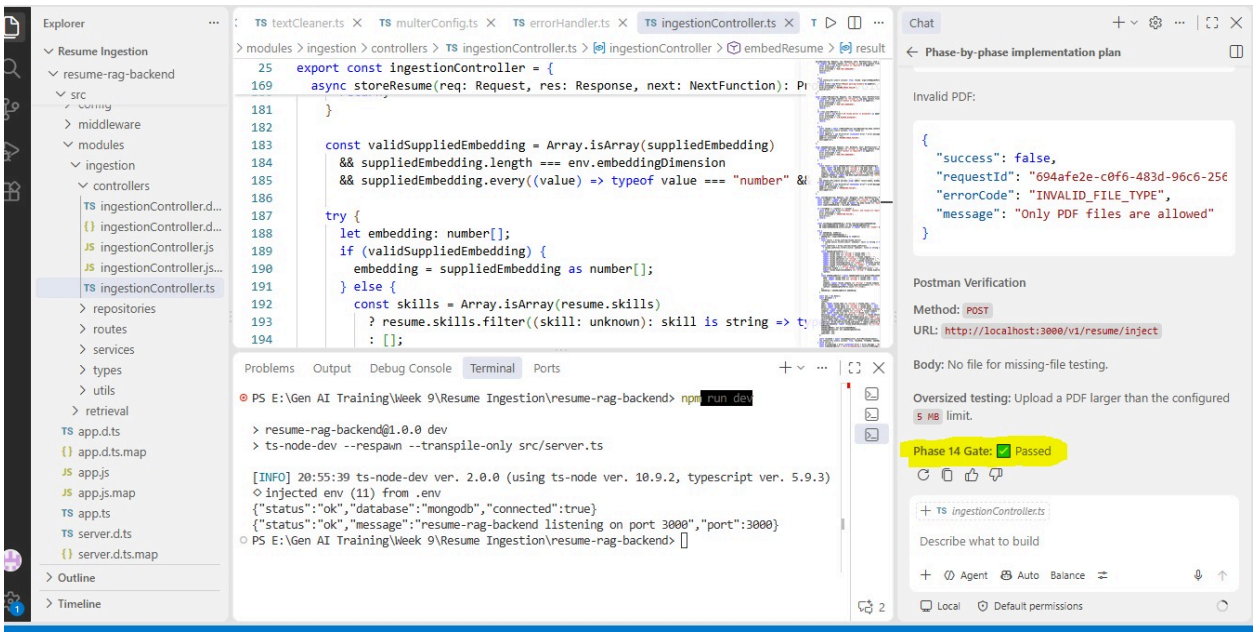

HR Application > Phase 12 Complete — MongoDB Resume Storage
POST
http://localhost:3000/v1/resume/store
Send
{
"fileName": "Rajesh Mohan Kumar_Test Architect.pdf",
"resume": {
"name": "Rajesh Mohan Kumar",
"role": "Test Architect & Senior Agentic Test Engineer",
"company": "Testleaf Software Solutions Private Limited",
"education": "B.Tech - Information Technology",
"totalExperience": 13,
"relevantExperience": null,
"skills": [
"Selenium WebDriver",
200 OK
{
"success": true,
"resumeId": "6ae7614ece13574e804f311e",
"fileName": "Rajesh Mohan Kumar_Test Architect.pdf",
"embeddingDimension": 1024,
"message": "Resume stored successfully"
}

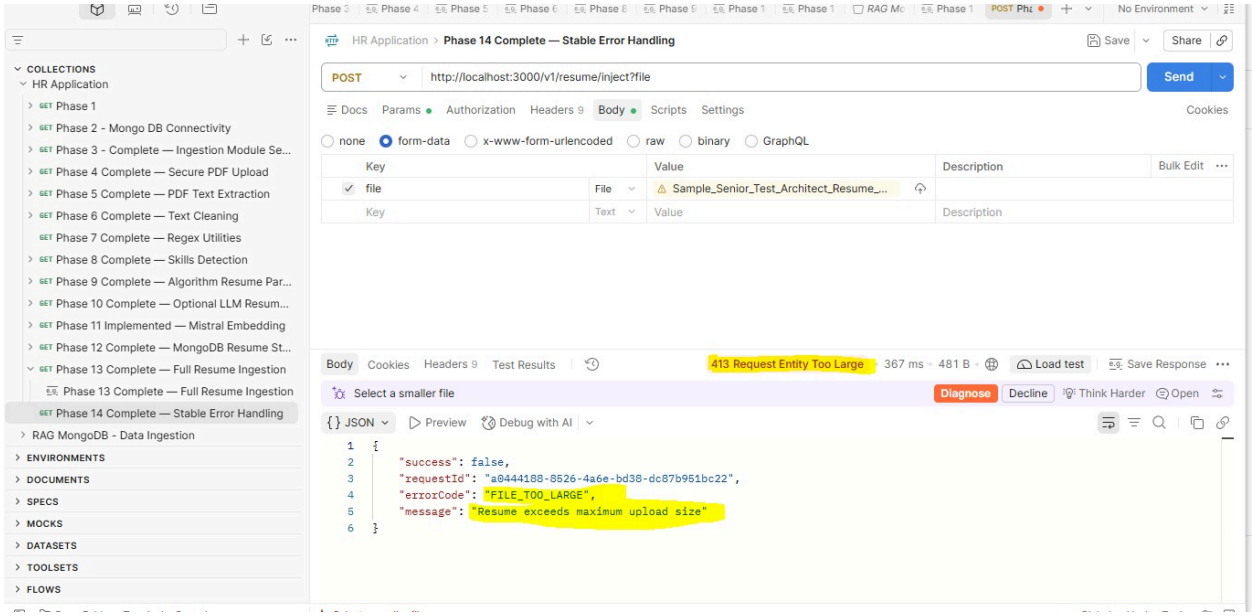
Phase 13:


Phase-by-phase implementation plan
Model is mistral-embed
Dimension is 1024
Postman Test
Method: POST
URL:
http://localhost:3000/v1/resume/inject
Body: form-data
Key: file
Type: File
Value: Select a resume PDF
Expected fields: success, message, resumeId, parsed, data, timing metrics.
Phase 13 Gate: Passed
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend> npm run dev
> resume-rag-backend@1.0.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts
[INFO] 20:55:39 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typescript ver. 5.9.3)
injected env (11) from .env
{"status":"ok","database":"mongodb","connected":true}
{"status":"ok","message":"resume-rag-backend listening on port 3000","port":3000}
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend>

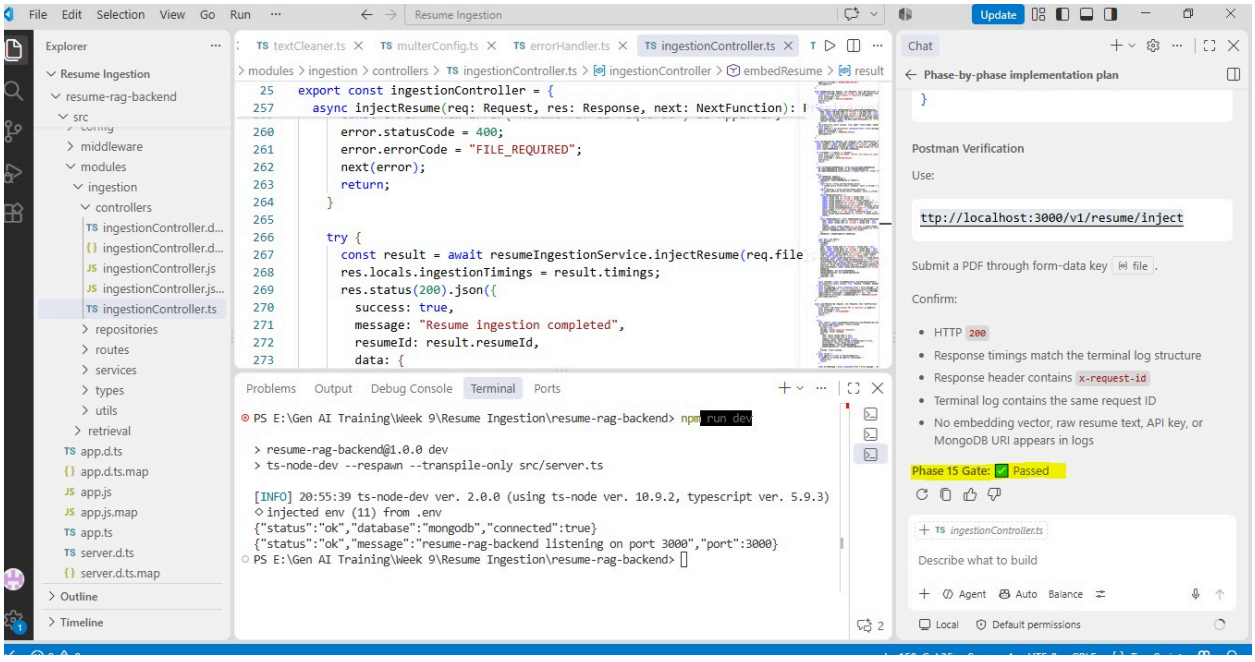

HR Application > Phase 13 Complete — Full Resume Ingestion
POST
http://localhost:3000/v1/resume/inject
Send
Save
Share
Docs Params Authorization Headers 9 Body Scripts Settings
none form-data x-www-form-urlencoded raw binary GraphQL
Key Value Description Bulk Edit
file File Arishraj Resume (1).pdf
Body Cookies Headers 9 Test Results
200 OK 3.64 s 707 B Load test Save Response
{ } JSON Preview Visualize
{
"success": true,
"message": "Resume ingestion completed",
"resumeId": "6ae7643b696fa4731ab00cab",
"data": {
"name": "ARISHRAJ T",
"role": "Finance Admin Intern",
"company": "Mani Power Tools",
"totalExperience": 2,
"skillsCount": 15,
"embeddingModel": "mistral-embed",
"embeddingDimension": 1024
COLLECTIONS
GenAI-Sep
HR Application
Phase 1
Phase 2 - Mongo DB Connectivity
Phase 3 - Complete — Ingestion Module Set...
Phase 4 Complete — Secure PDF Upload
Phase 5 Complete — PDF Text Extraction
Phase 6 Complete — Text Cleaning
Phase 7 Complete — Regex Utilities
Phase 8 Complete — Skills Detection
Phase 9 Complete — Algorithm Resume Pars...
Phase 10 Complete — Optional LLM Resume ...
Phase 11 Implemented — Mistral Embedding
Phase 12 Complete — MongoDB Resume Sto...
Phase 13 Complete — Full Resume Ingestion
RAG MongoDB - Data Ingestion
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Open Folder Terminal Console
Ready to go
Globals Vault Tools

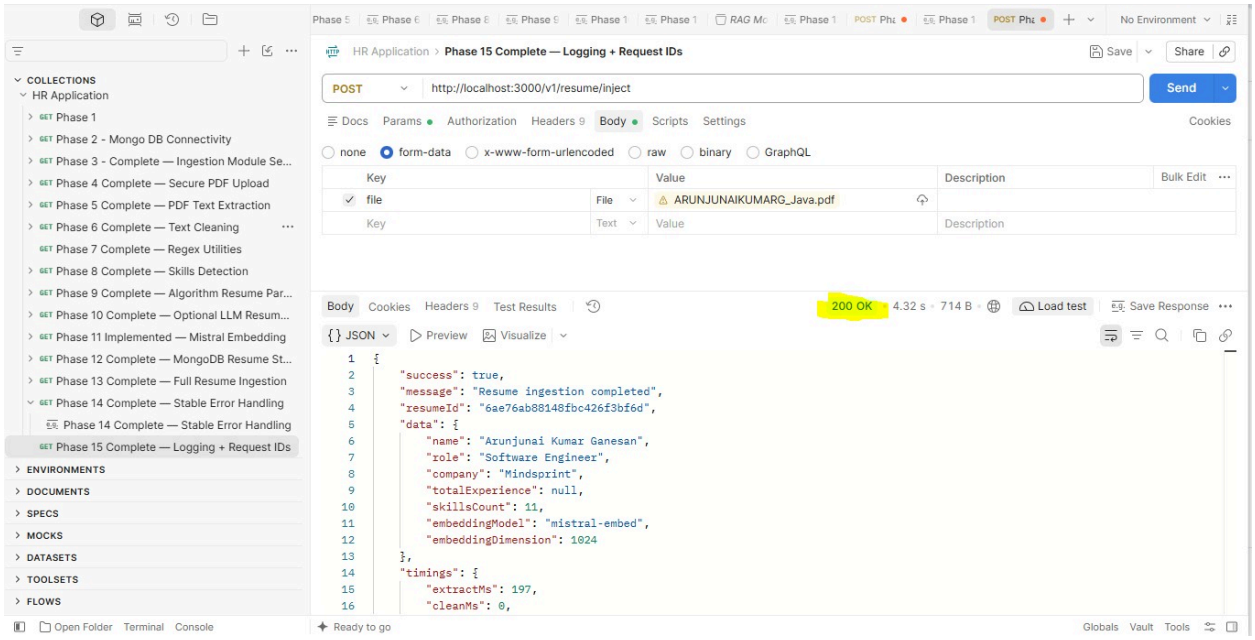
Phase 14:


Explorer
Resume Ingestion
resume-rag-backend
src
middleware
modules
ingestion
controllers
ingestionController.d...
ingestionController.d...
ingestionController.js
ingestionController.js...
ingestionController.ts
repositories
routes
services
types
utils
retrieval
app.d.ts
app.d.ts.map
app.js
app.js.map
app.ts
server.d.ts
server.d.ts.map
Outline
Timeline
textCleaner.ts multerConfig.ts errorHandler.ts ingestionController.ts
modules > ingestion > controllers > ingestionController.ts > ingestionController > embedResume > result
export const ingestionController = {
async storeResume(req: Request, res: Response, next: NextFunction): Pr
}
const validSuppliedEmbedding = Array.isArray(suppliedEmbedding)
&& suppliedEmbedding.length === env.embeddingDimension
&& suppliedEmbedding.every((value) => typeof value === "number" &&
try {
let embedding: number[];
if (validSuppliedEmbedding) {
embedding = suppliedEmbedding as number[];
} else {
const skills = Array.isArray(resume.skills)
? resume.skills.filter((skill: unknown): skill is string => ty
: [];
Problems Output Debug Console Terminal Ports
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend> npm run dev
> resume-rag-backend@1.0.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts
[INFO] 20:55:39 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typescript ver. 5.9.3)
injected env (11) from .env
{"status":"ok","database":"mongodb","connected":true}
{"status":"ok","message":"resume-rag-backend listening on port 3000","port":3000}
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend>
Chat
Phase-by-phase implementation plan
Invalid PDF:
{
"success": false,
"requestId": "694afe2e-c0f6-483d-96c6-256
"errorCode": "INVALID_FILE_TYPE",
"message": "Only PDF files are allowed"
}
Postman Verification
Method: POST
URL: http://localhost:3000/v1/resume/inject
Body: No file for missing-file testing.
Oversized testing: Upload a PDF larger than the configured 5 MB limit.
Phase 14 Gate: Passed
ingestionController.ts
Describe what to build
Agent Auto Balance
Local Default permissions

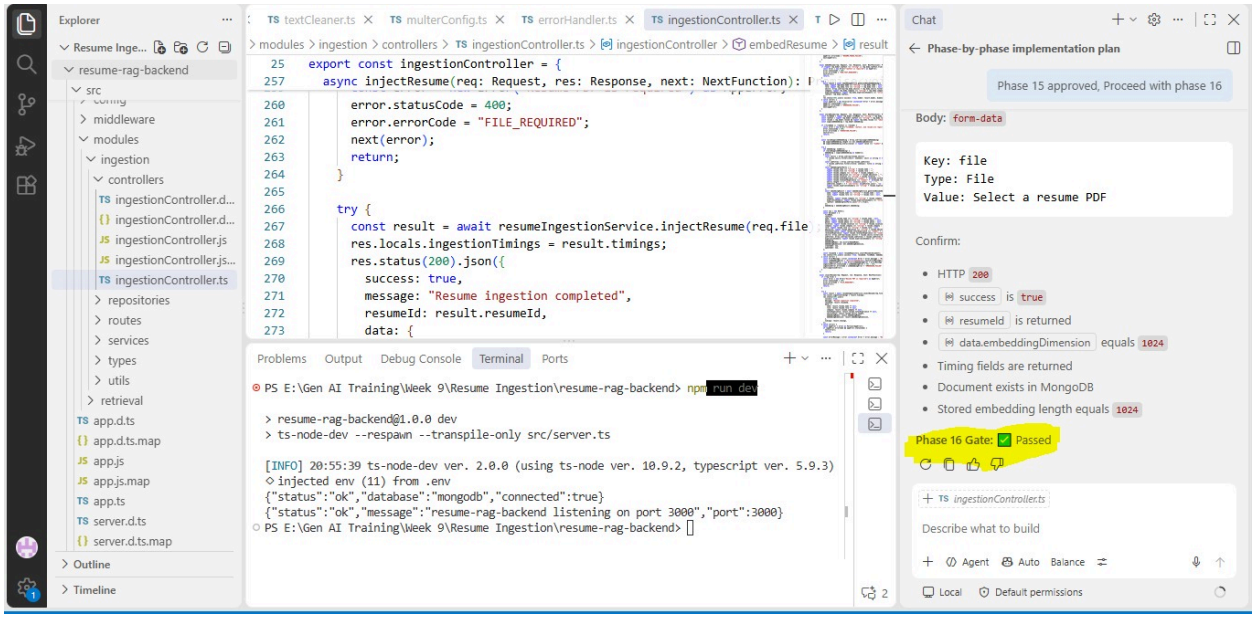

HR Application > Phase 14 Complete — Stable Error Handling
POST
http://localhost:3000/v1/resume/inject?file
Send
Save
Share
Docs Params Authorization Headers 9 Body Scripts Settings Cookies
none form-data x-www-form-urlencoded raw binary GraphQL
Key Value Description Bulk Edit
file File Sample_Senior_Test_Architect_Resume_...
Body Cookies Headers 9 Test Results
413 Request Entity Too Large
367 ms 481 B
Load test
Save Response
Select a smaller file
Diagnose Decline Think Harder Open
JSON Preview Debug with AI
{
"success": false,
"requestId": "a0444188-8526-4a6e-bd38-dc87b951bc22",
"errorCode": "FILE_TOO_LARGE",
"message": "Resume exceeds maximum upload size"
}

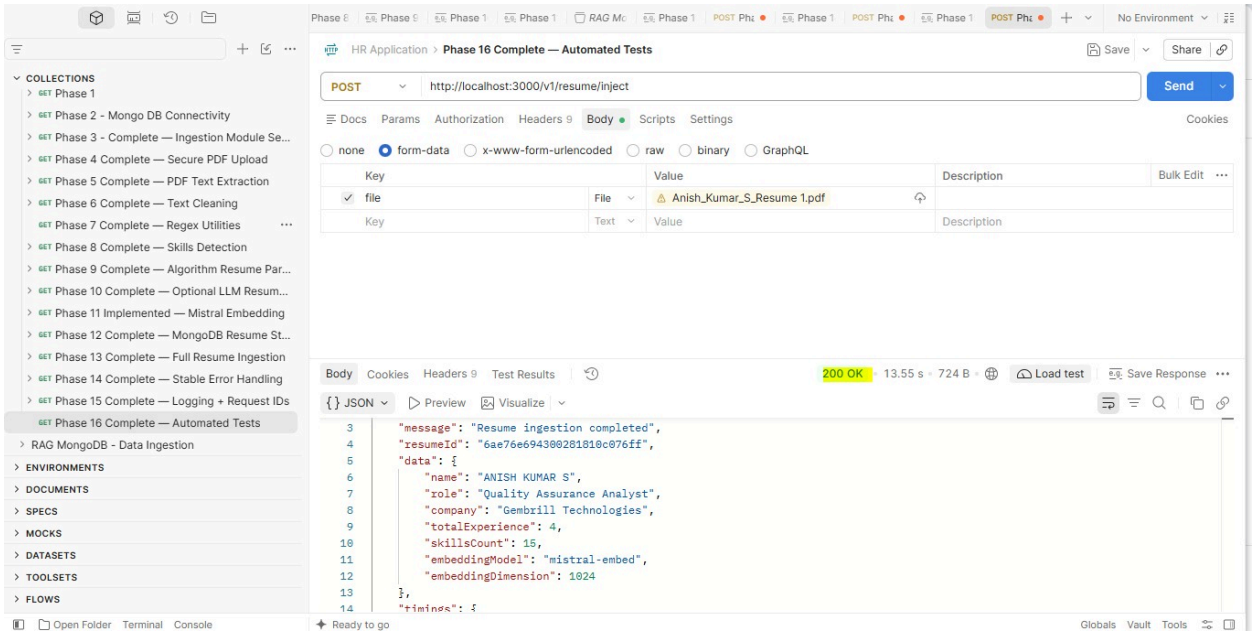
Phase 15:


Resume Ingestion
Update
Explorer
Resume Ingestion
resume-rag-backend
src
middleware
modules
ingestion
controllers
ingestionController.ts
repositories
routes
services
types
utils
retrieval
app.d.ts
app.js
app.ts
server.d.ts
Outline
Timeline
textCleaner.ts multerConfig.ts errorHandler.ts ingestionController.ts
modules > ingestion > controllers > ingestionController.ts > ingestionController > embedResume > result
export const ingestionController = {
async injectResume(req: Request, res: Response, next: NextFunction):
error.statusCode = 400;
error.errorCode = "FILE_REQUIRED";
next(error);
return;
try {
const result = await resumeIngestionService.injectResume(req.file
res.locals.ingestionTimings = result.timings;
res.status(200).json({
success: true,
message: "Resume ingestion completed",
resumeId: result.resumeId,
data: {
Problems Output Debug Console Terminal Ports
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend> npm run dev
> resume-rag-backend@1.0.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts
[INFO] 20:55:39 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typescript ver. 5.9.3)
injected env (11) from .env
{"status":"ok","database":"mongodb","connected":true}
{"status":"ok","message":"resume-rag-backend listening on port 3000","port":3000}
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend>
Chat
Phase-by-phase implementation plan
Postman Verification
Use:
ttp://localhost:3000/v1/resume/inject
Submit a PDF through form-data key file.
Confirm:
HTTP 200
Response timings match the terminal log structure
Response header contains x-request-id
Terminal log contains the same request ID
No embedding vector, raw resume text, API key, or MongoDB URI appears in logs
Phase 15 Gate: Passed
ingestionController.ts
Describe what to build
Agent Auto Balance
Local Default permissions


HR Application
Phase 15 Complete — Logging + Request IDs
Save
Share
POST
http://localhost:3000/v1/resume/inject
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Key
Value
Description
Bulk Edit
file
File
ARUNJUNAIKUMARG_Java.pdf
Key
Text
Value
Description
Body
Cookies
Headers 9
Test Results
200 OK
4.32 s
714 B
Load test
Save Response
{} JSON
Preview
Visualize
{
"success": true,
"message": "Resume ingestion completed",
"resumeId": "6ae76ab88148fbc426f3bf6d",
"data": {
"name": "Arunjunai Kumar Ganesan",
"role": "Software Engineer",
"company": "Mindsprint",
"totalExperience": null,
"skillsCount": 11,
"embeddingModel": "mistral-embed",
"embeddingDimension": 1024
},
"timings": {
"extractMs": 197,
"cleanMs": 0,
COLLECTIONS
HR Application
Phase 1
Phase 2 - Mongo DB Connectivity
Phase 3 - Complete — Ingestion Module Se...
Phase 4 Complete — Secure PDF Upload
Phase 5 Complete — PDF Text Extraction
Phase 6 Complete — Text Cleaning
Phase 7 Complete — Regex Utilities
Phase 8 Complete — Skills Detection
Phase 9 Complete — Algorithm Resume Par...
Phase 10 Complete — Optional LLM Resum...
Phase 11 Implemented — Mistral Embedding
Phase 12 Complete — MongoDB Resume St...
Phase 13 Complete — Full Resume Ingestion
Phase 14 Complete — Stable Error Handling
Phase 14 Complete — Stable Error Handling
Phase 15 Complete — Logging + Request IDs
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
Open Folder
Terminal
Console
Ready to go
Globals
Vault
Tools

Phase 16:


Explorer
Resume Inge...
resume-rag-backend
src
middleware
modules
ingestion
controllers
ingestionController.d...
ingestionController.d...
ingestionController.js
ingestionController.js...
ingestionController.ts
repositories
routes
services
types
utils
retrieval
app.d.ts
app.d.ts.map
app.js
app.js.map
app.ts
server.d.ts
server.d.ts.map
Outline
Timeline
textCleaner.ts
multerConfig.ts
errorHandler.ts
ingestionController.ts
modules > ingestion > controllers > ingestionController.ts > ingestionController > embedResume > result
25 export const ingestionController = {
257 async injectResume(req: Request, res: Response, next: NextFunction): P
260 error.statusCode = 400;
261 error.errorCode = "FILE_REQUIRED";
262 next(error);
263 return;
264 }
265
266 try {
267 const result = await resumeIngestionService.injectResume(req.file
268 res.locals.ingestionTimings = result.timings;
269 res.status(200).json({
270 success: true,
271 message: "Resume ingestion completed",
272 resumeId: result.resumeId,
273 data: {
Problems
Output
Debug Console
Terminal
Ports
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend> npm run dev
> resume-rag-backend@1.0.0 dev
> ts-node-dev --respawn --transpile-only src/server.ts
[INFO] 20:55:39 ts-node-dev ver. 2.0.0 (using ts-node ver. 10.9.2, typescript ver. 5.9.3)
injected env (11) from .env
{"status":"ok","database":"mongodb","connected":true}
{"status":"ok","message":"resume-rag-backend listening on port 3000","port":3000}
PS E:\Gen AI Training\Week 9\Resume Ingestion\resume-rag-backend>
Chat
Phase-by-phase implementation plan
Phase 15 approved, Proceed with phase 16
Body: form-data
Key: file
Type: File
Value: Select a resume PDF
Confirm:
HTTP 200
success is true
resumeId is returned
data.embeddingDimension equals 1024
Timing fields are returned
Document exists in MongoDB
Stored embedding length equals 1024
Phase 16 Gate: Passed
ingestionController.ts
Describe what to build
Agent
Auto
Balance
Local
Default permissions


COLLECTIONS
Phase 1
Phase 2 - Mongo DB Connectivity
Phase 3 - Complete — Ingestion Module Se...
Phase 4 Complete — Secure PDF Upload
Phase 5 Complete — PDF Text Extraction
Phase 6 Complete — Text Cleaning
Phase 7 Complete — Regex Utilities
Phase 8 Complete — Skills Detection
Phase 9 Complete — Algorithm Resume Par...
Phase 10 Complete — Optional LLM Resum...
Phase 11 Implemented — Mistral Embedding
Phase 12 Complete — MongoDB Resume St...
Phase 13 Complete — Full Resume Ingestion
Phase 14 Complete — Stable Error Handling
Phase 15 Complete — Logging + Request IDs
Phase 16 Complete — Automated Tests
RAG MongoDB - Data Ingestion
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
HR Application > Phase 16 Complete — Automated Tests
Save
Share
POST
http://localhost:3000/v1/resume/inject
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Key
Value
Description
Bulk Edit
file
File
Anish_Kumar_S_Resume 1.pdf
Text
Body
Cookies
Headers 9
Test Results
200 OK
13.55 s
724 B
Load test
Save Response
JSON
Preview
Visualize
"message": "Resume ingestion completed",
"resumeId": "6ae76e694300281810c076ff",
"data": {
"name": "ANISH KUMAR S",
"role": "Quality Assurance Analyst",
"company": "Gembrill Technologies",
"totalExperience": 4,
"skillsCount": 15,
"embeddingModel": "mistral-embed",
"embeddingDimension": 1024
},
Open Folder
Terminal
Console
Ready to go
Globals
Vault
Tools